Replication package for "Rent Control? Bad in Theory, Not Always in Practice"

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Shahar Rotberg and Lin Zhang

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Repository: https://github.com/srotberg/rent_control_bad_in_theory_not_always_in_practice

Contact: Shahar Rotberg, shahar.rotberg@gmail.com

1. Overview

The paper builds a quantitative life-cycle model of the Toronto housing market with tenancy rent control, calibrates it to Canadian data, and compares the benchmark with four counterfactual policies. This package contains:

- `code/`: the Fortran source for the model, which produces every simulation result in the paper, and instructions for running it (`code/readme.txt`).
- `data/`: the data and Stata scripts behind the calibration targets and the data moments in Tables 1-4 and Figure 2.
- `outputs/`: the output file of every simulation, including the six that the paper reports.
- `plots/`: the Python script that draws Figure 2, and the figure it produces.

The package is about 18 MB. Section 6 lists every file.

The paper's tables are typed from the simulation output files; Section 5 explains, for each table cell, which output line it comes from. Figure 1 is a conceptual diagram and uses no data.

2. Data availability and provenance

All data files included in this package may be redistributed.

Source	Used for	In this package
2016 Census of Population, confidential microdata (Statistics Canada)	Toronto CMA moments: homeownership by age, household size, income quintiles, Gini, income taxes, rent-to-income distribution, shares of renters in controlled units	Vetted results only: `data/2016_Census_vetted.xlsx`. Scripts: `Census_2016.do`, `proportions_of_renters_market_and_quintiles_2016.do`, `rent_to_income_cdf_2016.do`
2016 Census Hierarchical Public Use Microdata File (Statistics Canada)	Controlled-rent entry premium, discrimination tests, shares of new and old rental units (Appendix A)	Yes: `data/census_pumf_clean_2016.dta` and the user guide. Scripts: `premium_and_discrimination.do`, `computations_fraction_controlled_new_and_old.do`

Source	Used for	In this package
2019 Survey of Financial Security (Statistics Canada)	LTV by age, mortgage debt to house value, share of owners with a mortgage, wealth distribution, net worth to GDP, bequests	Results: `data/2019_SFS.xlsx`. Scripts: `SFS_2019.do`, `net_worth_to_GDP.do`
CMHC Rental Market Survey	Toronto average rents by bedroom type	Yes: `data/CMHC_RMS.csv`
World Health Organization life tables for Canada	Death rates used to compute bequests	Yes: `data/WHO_death_rates_Canada.xlsx` (source URL inside the file)
5-year variable mortgage rates and 10-year bond yields, 2002-2017	Mortgage interest rate premium (Table 1)	Yes: `data/bond_yields_mortgage_rates.csv`
Published estimates	Parameters cited in Table 1 (e.g., Guvenen et al. 2023; Gruber and Martin 2003)	Values set in the Fortran code; references in the paper

Confidential Census microdata. The authors computed the statistics in `data/2016_Census_vetted.xlsx` at the Canada Mortgage and Housing Corporation from the confidential 2016 Census microdata, using the three Census scripts listed above; the results were vetted for release. The microdata file (`Census_2016.dta`) is confidential and is not included. Researchers can apply to Statistics Canada for access; the scripts are included so the computations can be rerun by anyone with access.

SFS microdata. The 2019 SFS microdata file (`SFS_2019.dta`) is distributed by Statistics Canada and is not included. The statistics computed from it are in `data/2019_SFS.xlsx`.

3. Computational requirements

- **Model (Fortran).** gfortran with OpenMP; no other libraries. The code compiles with GNU Fortran 13.3 on Linux (Ubuntu 24.04) using the command in `code/readme.txt`. The paper's simulations were run on a multi-core workstation.
- **Data scripts (Stata).** The `.do` files use standard Stata commands (`svy`, `xtile`, `egen`). They need the microdata files described in Section 2.
- **Figure 2 (Python).** Python 3 with `matplotlib` and `openpyxl` (tested with Python 3.13, matplotlib 3.10, openpyxl 3.1).

Run times. Each output file records its run time on its second line (`hours to finish simulaion`). For the simulations reported in the paper:

Simulation	Run time
Benchmark	7.7 hours
Experiment 1	8.5 hours
Experiment 2	5.7 hours
Experiment 3	5.5 hours
Experiment 4 (one discount value)	168 hours (about 7 days)
Experiment 4 with income conditioning	155 hours (about 6.5 days)

The full search over the Experiment 4 discount took between 1.5 and 7 days per grid point. Figure 2 and the Stata scripts each run in under a minute.

4. Instructions

1. **Simulations.** Follow `code/readme.txt`. Run the benchmark first, then the counterfactuals in the same folder, because every counterfactual reads the benchmark's `.dat` files. Each run writes `outputs<simulation name>.txt`; the six that the paper uses are listed in Section 5.
2. **Tables 2-6.** Read the values from the output files as described in Section 5.
3. **Figure 2.** From the package root, run `python plots/figure_2_homeownership_and_LTV.py`. It reads `data/2016_Census_vetted.xlsx`, `data/2019_SFS.xlsx` and the benchmark output file, and writes `plots/Figure_2_homeownership_and_LTV.png` and `.tiff`.
4. **Data moments.** Each Stata script loads its microdata with a `use` command near the top, preceded by a `cd` or `log using` line with the authors' local paths; change those paths to where the microdata are stored, then run the script. The results are the values reported in `2016_Census_vetted.xlsx` and `2019_SFS.xlsx`.

5. From output files to tables and figures

The six simulations reported in the paper (all files are in `outputs/`):

Simulation	Output file
Benchmark (tenancy control)	`outputs_control_sorting_benchmark_marginal_profits.txt`
Experiment 1: decontrol, income tax cuts	`outputs0_control_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt`
Experiment 2: decontrol, transfers to renters and the homeless	`outputs0_control_GE_counter_marginal_pi_lump_sum_renters.txt`
Experiment 3: decontrol, full profit tax, transfers to newborns	`outputs0_control_GE_counter_marginal_pi_lump_sum_young_pi_full_tax.txt`
Experiment 4: optimal unit control	`outputs28_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt`
Experiment 4, controlled units for low-income renters only	`outputs28_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top_controlled_rentals_to_low_income_only.txt`

Each output file is a list of lines of the form `name value`. In what follows, "line `x`" means the value on the line whose name is `x` (the first such line unless stated otherwise).

Table 1 (externally assigned parameters). The values are set in the Fortran code. Sources are listed in the table and in Appendix A; the data-based values come from the files in `data/` (Section 2).

Table 2 (internally calibrated parameters). Target moments: `2016_Census_vetted.xlsx`, `2019_SFS.xlsx`, Zhang and Guo (2022) and the Homelessness Hub, as stated in the table notes. Model moments, from the benchmark output:

Moment	Output line
Mortgages / housing value	`model mortgage debt to gross housing wealth`
Average rent / income	`model_average_rent_to_income_ratio`
Total bequests / total net wealth	`model_bequests/model_net_wealth`
Average income tax rate	`model_average_labour_income_tax`
Labor income Gini	`gini_coefficient_income`
Average controlled rent / free-market rent	second `model_average_rent` line divided by the first
Homeownership rate	`model_homeownership_rate`
Renters with rent/income above 30%	1 minus `rent_to_income_cdf_30`

Moment	Output line
Homelessness rate (%)	`model_fraction_homeless (%)`

Table 3 (non-targeted statistics). Data: `2019_SFS.xlsx`, `2016_Census_vetted.xlsx`, `net_worth_to_GDP.do` and Chatterjee and Eyigungor (2015). Model, from the benchmark output:

Statistic	Output line
House price to GDP per household	`housing_price(j)*model_average_owner_occupied_unit_size(j)/(model_GDP_incl_imput_rent(j)/total_population)`
Owners with a mortgage	`model_fraction_of_homeowners_with_mortgage`
Rental-to-owner unit value	`model_average_rental_unit_size(j)/model_average_owner_occupied_unit_size(j)`
Net wealth to GDP	`model total net wealth to output`
Renters in the free market	`model_fraction_living_in_free`
Renters spending over 50% / 75% of income on rent	1 minus `rent_to_income_cdf_50` / `rent_to_income_cdf_75`
Wealth shares of the top 1%, 10%, 20%	`model_wealth_of_top_x_perc(k)/model_net_wealth` for k = 1, 2, 3
Wealth share of the bottom 50%	1 minus the same line for k = 4
Wealth Gini	`gini_coefficient`

Table 4 (households in the controlled market by income quintile). Data row: `2016_Census_vetted.xlsx`, sheet `rent_control` (`proportions_of_renters_market_and_quintiles_2016.do`). Model rows: lines `market_1_quintile_1` to `market_1_quintile_5` in the benchmark and Experiment 4 output files.

Table 5 (aggregate effects). Each row compares an experiment's output file with the benchmark's:

Column	Computation
Newborn welfare (%)	$100\,[(W_1/W_0)^{1/((1-\gamma)(1-\sigma))} - 1]$, where W is line `average_welfare_of_youngest_cohort`, $\gamma = 0.385$ and $\sigma = 2$ (Section 5.2 of the paper)
Controlled initial rent (%)	second `rent/rent_calibrated` line minus 1 (Experiment 4 only)
Free-market rent (%)	first value on line `model_average_rent(:)/average_rent_calibrated(:)` minus 1
Rental quantity (%)	third `total_rental_supply` line, relative to the benchmark
Housing price (%)	`housing_price/calibrated_stationary_housing_price` minus 1
Homeownership (p.p.)	difference in `model_homeownership_rate`, times 100
Homelessness (%)	ratio of `model_fraction_homeless (%)` to the benchmark's, minus 1

Table 6 (welfare by income quintile). The welfare formula above applied to lines `welfare_of_quintile 1` to `welfare_of_quintile 5`.

Values in the paper are rounded to two decimals; a few cells differ from the output files by 0.01 because of rounding.

Figure 1. Conceptual diagram drawn by the authors; no data or code.

Figure 2. `plots/figure_2_homeownership_and_LTV.py`. Data: `2016_Census_vetted.xlsx` (sheet `homeownership_housholds_szie`) and `2019_SFS.xlsx` (sheet `LTV by Age Group`). Model: lines `homeownership_age1` to `homeownership_age60` (ages 26-85) and `LTV_age_group1` to `LTV_age_group12` in the benchmark output.

Numbers in the text.

- Experiment 4's discount of 35.6% relative to the benchmark free-market rent: the second `model_average_rent` line of the Experiment 4 output divided by the first `model_average_rent` line of the benchmark output.
- The 2.57% welfare gain with income conditioning (footnote in Section 5.1): the welfare formula applied to the income-conditioned Experiment 4 output.
- The optimal discount: Experiment 4 was run for a grid of values of `discount_in_unit_control` (files `outputsNN_discount_...`; see `code/readme.txt`). The run labelled 28 (`discount_in_unit_control=0.27`) gives the highest newborn welfare.

6. List of all files

code/

- `code/readme.txt` (5 KB). How to compile and run the model and which switches select each simulation.
- `code/rent_control_bad_in_theory_not_always_in_practice_Rotberg_Zhang_v11.f90` (1.1 MB). Fortran source for the model; produces every simulation (benchmark and counterfactuals).

data/

- `data/2016_Census_Hierarchical_PUMF_User_Guide.pdf` (3.3 MB). Statistics Canada user guide for the 2016 Census Hierarchical PUMF.
- `data/2016_Census_vetted.xlsx` (18 KB). Vetted statistics from the confidential 2016 Census microdata (Toronto CMA): rent control shares by income quintile, rent-to-income distribution, income and Gini, homeownership and household size by age.
- `data/2019_SFS.xlsx` (23 KB). Statistics computed from the 2019 Survey of Financial Security: LTV by age group, mortgage debt to house value, wealth distribution, bequests.
- `data/CMHC_RMS.csv` (4 KB). CMHC Rental Market Survey: Toronto average rents by bedroom type, 1990-2024.
- `data/Census_2016.do` (16 KB). Stata: homeownership by age, household size, income quintiles, Gini, income taxes, controlled vs. uncontrolled renters (confidential 2016 Census microdata; results in `2016_Census_vetted.xlsx`).
- `data/SFS_2019.do` (3 KB). Stata: LTV by age, mortgage debt, house values, wealth shares, share of owners with a mortgage (2019 SFS microdata; results in `2019_SFS.xlsx`).
- `data/WHO_death_rates_Canada.xlsx` (16 KB). WHO life-table death rates for Canada; used to compute bequests (`2019_SFS.xlsx`, sheet "Wealth Bequest").
- `data/bond_yields_mortgage_rates.csv` (31 KB). 5-year variable mortgage rates and 10-year bond yields, 2002-2017; average spread gives the mortgage premium in Table 1.
- `data/census_pumf_clean_2016.dta` (9.6 MB). 2016 Census Hierarchical Public Use Microdata File (cleaned), Stata format.
- `data/computations_fraction_controlled_new_and_old.do` (1 KB). Stata (uses `census_pumf_clean_2016.dta`): shares of new and old rental units, condo vs. single-family stock.
- `data/net_worth_to_GDP.do` (0 KB). Stata: aggregate net worth to GDP (2019 SFS microdata).
- `data/premium_and_discrimination.do` (2 KB). Stata (uses `census_pumf_clean_2016.dta`): controlled-rent entry premium and age/income discrimination (Appendix A).
- `data/proportions_of_renters_market_and_quintiles_2016.do` (1 KB). Stata: share of households in controlled and free-market rentals by income quintile (Table 4, data row).
- `data/rent_to_income_cdf_2016.do` (2 KB). Stata: distribution of renters' rent-to-income ratios (Tables 2 and 3).

outputs/

- outputs/outputs0_control_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt (58 KB). Experiment 1: decontrol with income tax cuts (Tables 5-6, row 1).
- outputs/outputs0_control_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top_higher_elasticity.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (higher rental supply elasticity).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_bottom_renters.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (transfers to bottom-quintile renters).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_renters.txt (58 KB). Experiment 2: decontrol with transfers to renters and the homeless (Tables 5-6, row 2).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_renters_higher_elasticity.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (higher rental supply elasticity).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_to_all_by_income_and_age.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (transfers to all households by income and age).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_two_bottom_renters.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (transfers to bottom-two-quintile renters).
- outputs/outputs0_control_GE_counter_marginal_pi_lump_sum_young_pi_full_tax.txt (58 KB). Experiment 3: decontrol, full profit tax, transfers to newborns (Tables 5-6, row 3).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_inc_tax_reduction_pi_to_top_sr.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_inc_tax_reduction_pi_to_top_sr_fixed_stocks.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (housing stocks held fixed; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_inc_tax_reduction_pi_to_top_sr_fixed_stocks_relative_to_unit_control.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (measured against the unit-control economy; housing stocks held fixed; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_inc_tax_reduction_pi_to_top_sr_relative_to_unit_control.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (measured against the unit-control economy; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_renters_sr.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_renters_sr_fixed_stocks.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (housing stocks held fixed; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_renters_sr_fixed_stocks_relative_to_unit_control.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (measured against the unit-control economy; housing stocks held fixed; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_renters_sr_relative_to_unit_control.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (measured against the unit-control economy; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_young_pi_full_tax_sr.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_young_pi_full_tax_sr_fixed_stocks.txt (58 KB). Additional simulation from an earlier revision, not reported in the final paper (housing stocks held fixed; short run; bequests held fixed).
- outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_young_pi_full_tax_sr_fixed_stocks_relative_to_unit_control.txt (58 KB). Additional simulation from an earlier

revision, not reported in the final paper (measured against the unit-control economy; housing stocks held fixed; short run; bequests held fixed).

- `outputs/outputs0_control_beq_constant_counter_marginal_pi_lump_sum_young_pi_full_tax_sr_relative_to_unit_control.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (measured against the unit-control economy; short run; bequests held fixed).
- `outputs/outputs0_control_rents_and_hp_constant_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (rents and house prices held at benchmark values).
- `outputs/outputs0_control_rents_and_hp_constant_counter_marginal_pi_lump_sum_renters.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (rents and house prices held at benchmark values).
- `outputs/outputs0_control_rents_and_hp_constant_counter_marginal_pi_lump_sum_young_pi_full_tax.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (rents and house prices held at benchmark values).
- `outputs/outputs16_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 16); used to choose the welfare-maximizing discount.
- `outputs/outputs24_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 24); used to choose the welfare-maximizing discount.
- `outputs/outputs25_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 25); used to choose the welfare-maximizing discount.
- `outputs/outputs26_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 26); used to choose the welfare-maximizing discount.
- `outputs/outputs28_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4: optimal unit control, discount_in_unit_control=0.27 (Tables 4-6).
- `outputs/outputs28_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top_controlled_rentals_to_low_income_only.txt` (58 KB). Experiment 4 with controlled units for low-income renters only (2.57% welfare gain, footnote in Section 5.1).
- `outputs/outputs29_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 29); used to choose the welfare-maximizing discount.
- `outputs/outputs30_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 30); used to choose the welfare-maximizing discount.
- `outputs/outputs31_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 31); used to choose the welfare-maximizing discount.
- `outputs/outputs32_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Experiment 4 grid point (discount label 32); used to choose the welfare-maximizing discount.
- `outputs/outputs_0_control_sorting_benchmark_marginal_profits_higher_elasticity.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (higher rental supply elasticity).
- `outputs/outputs_25_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Separate run of a discount grid point (not used in the paper); the space the code wrote into the file name is replaced by an underscore.
- `outputs/outputs_28_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Separate run of a discount grid point (not used in the paper); the space the code wrote into the file name is replaced by an underscore.
- `outputs/outputs_31_discount_control_random_GE_counter_marginal_pi_inc_tax_reduction_pi_to_top.txt` (58 KB). Separate run of a discount grid point (not used in the paper); the space the code wrote into the file name is replaced by an underscore.

- `outputs/outputs_control_sorting_benchmark_marginal_profits.txt` (58 KB). Benchmark (tenancy control): Tables 2-4 and Figure 2 (model); baseline for Tables 5-6.
- `outputs/outputs_control_sorting_benchmark_marginal_profits_higher_elasticity.txt` (58 KB). Additional simulation from an earlier revision, not reported in the final paper (higher rental supply elasticity).

plots/

- `plots/Figure_2_homeownership_and_LTV.png` (270 KB). Figure 2 (300 dpi), produced by the script above.
- `plots/Figure_2_homeownership_and_LTV.tiff` (1.4 MB). Figure 2 (600 dpi TIFF), produced by the script above.
- `plots/figure_2_homeownership_and_LTV.py` (5 KB). Python: draws Figure 2 from data/ and the benchmark output file.

The GitHub repository also contains `writing/`, the LaTeX source of the paper; it is not part of the replication package.